

Embodied versus operational emissions: A life cycle optimization framework for building envelope design under climate change

Zhuocheng Duan^{*}, Hossein Omrany, Jian Zuo

School of Architecture and Civil Engineering, The University of Adelaide, Australia

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ABSTRACT

Buildings contribute over one-third of global energy-related greenhouse gas (GHG) emissions, with embodied emissions gaining importance as operational efficiency improves. However, the dynamic relationship between embodied and operational impacts under global warming remains insufficiently understood. Therefore, this research aims to address this gap by developing an integrated life cycle assessment (LCA) framework that decomposes embodied-operational emissions trade-offs in building envelope design across four distinct scenarios: S1) static baseline, S2) dynamic climate, S3) evolving energy grids, and S4) coupled climate-energy dynamics. This study employs multi-objective optimization to evaluate diverse building envelope design variants for a 10-storey mass timber office building across Australia. Optimization results reveal profound spatial heterogeneity, with minimum life cycle GHG emissions (LCGHGE) ranging six-fold from 1426 (Hobart) to 8322 kg CO₂eq/m² (Townsville). The framework reveals that climate change (S2) degrades performance by 2.46–9.69 % across most regions, with tropical cities experiencing the greatest vulnerability. Grid carbon intensity represents one of the most critical factors, as S3 demonstrates 28.32–38.37 % emission reductions in high-emission regions but approaches saturation limits in low-carbon grids. Most importantly, the coupled climate-energy scenario (S4) exposes complex interaction effects where global warming impacts partially offset grid decarbonization benefits by 2.38–8.00 % in most cities. Marginal trade-off rate analysis demonstrates that operational energy minimization beyond optimal points can inadvertently increase total carbon footprint. While demonstrated across Australian cities, the framework is replicable across other geographical contexts and building typologies, requiring locally appropriate climate scenarios, energy system emission factor trajectories, and material embodied GHG coefficients as inputs. By making the interactions between climate change, grid transition, and building performance explicit, the framework equips designers, engineers and policymakers with quantitative evidence to make more durable low-carbon design decisions.

1. Introduction

The building and construction sector is responsible for approximately 34 % of global energy and process-related global greenhouse gas (GHG) emissions in 2023 [1], reaching new highs as overall energy demand and emissions increased by approximately 1 % from 2021 [2]. According to the IEA [3], global floor area will rise by approximately 15 % by 2030 relative to 2022 levels, with 80 % expected to occur in developing and emerging economies. Such large amounts of emissions combined with the continued growth in global floor area raise concerns about whether the building sector is on track to achieve net zero emissions by 2050 [1]. Given the urgent global need to mitigate climate change, the building sector presents significant opportunities for

emissions reduction through the adoption of low-carbon construction materials and improved energy efficiency.

The building sector faces a complex dual challenge in reducing both operational and embodied impacts. While operational energy efficiency has been the primary focus of sustainability efforts, growing evidence suggests that embodied impacts arising from the manufacturing and processing of materials are becoming increasingly significant [4–6]. According to the UN report [1], embodied GHG emissions from materials contribute to 18 % of the building-related emissions. At the building level, embodied impacts can constitute up to 50 % of whole life carbon emissions in high-efficiency buildings and more than 90 % in extreme cases [7]. In fact, it is reported that some of the avoided operational energy consumption is shifted to the environmental impacts in

^{*} Corresponding authors.

E-mail address: zhuocheng.duan@adelaide.edu.au (Z. Duan).

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production and demolition stages [8]. Ramesh et al. [9] also revealed that excessive use of energy efficient measures in a building may be counterproductive. Such a shifting pattern necessitates an approach to balance operational and embodied impacts throughout the building lifespan.

Climate change is altering environmental conditions, meaning that buildings with operational lifespans exceeding 50 years will inevitably experience climate impacts and require adaptive design responses [10, 11]. The increased temperature, fluctuating humidity and intensified extremes directly impact building thermal performance [12]. However, historical climate data, which has long informed building standards and design practices, may no longer accurately predict future performance requirements [13]. The changing climate affects the balance between heating and cooling requirements, challenging conventional approaches to building envelope design.

Historical building codes and standards have prioritized reducing operational energy through stricter thermal performance requirements of the building envelope. While this approach has curbed energy use to some degree through measures like increased insulation, it requires additional materials and inadvertently increases embodied impacts. This trade-off creates a dilemma where over-insulation in pursuit of operational savings elevates upfront embodied emissions [14], which dominate the carbon footprint of high-performing buildings such as nearly zero energy buildings. Under-insulation, however, raises concerns regarding long-term operational emissions, a challenge that becomes particularly problematic as climate change alters future heating and cooling demands. This complexity can be further compounded when accounting for other variables such as regional variations in global warming impacts, local climate and building envelope design. This challenge highlights the need to examine building envelope design approaches that balance these competing impacts.

Contemporary building envelope optimization research demonstrates three interconnected methodological limitations. First, existing studies rely on static assessment approaches that fail to capture the temporal evolution of climate and energy systems simultaneously [15, 16]. Second, optimization frameworks treat embodied carbon as a fixed constraint or omit it entirely [17], despite its growing contribution to life cycle emissions in high-performance buildings. Third, research predominantly focuses on isolated envelope components, particularly wall insulation, while neglecting whole-system interactions among walls, roofs and windows [18]. No integrated framework has evaluated how simultaneous climate and energy transitions reshape the fundamental balance between embodied and operational carbon.

In this context, this study aims to address this gap by proposing a life cycle assessment (LCA) framework to identify the relationship between embodied and operational emissions under global warming. By integrating climate projections, material embodied GHG, and energy simulations, this work advances actionable strategies for resilient low-carbon building design in a warming world. Specifically, building designers and architects gain guidance for envelope designs that balance embodied and operational GHG under climate uncertainty. Policymakers and regulatory authorities can utilize these findings as evidence for updating building codes and standards to incorporate life cycle accounting and climate change adaptation requirements. Building owners and developers obtain decision-support frameworks for evaluating investment priorities across regional contexts, particularly identifying where envelope-driven interventions offer greatest decarbonization leverage. The research community benefits from a replicable methodological framework for investigating climate-energy-material system interdependencies, which is applicable to diverse building typologies, structural systems and different scales, e.g., individual buildings to district scale.

2. Literature review

2.1. Climate change impacts on building energy

Climate change significantly affects building energy performance through alterations in temperature patterns, solar radiation and extreme weather events [12]. A growing body of literature on the impact of climate change on the building sector has focused on 1) methods/tools to generate future weather files [10], 2) building thermal performance [11], and 3) adaptation and mitigation to the global warming [19,20]. This categorization reflects the progression from establishing climate projection tools, to understanding their implications for building energy consumption, to developing intervention strategies that respond to these changes.

2.1.1. Future weather file generation methods and tools

Future weather files enable designers and researchers to evaluate building performance under future climate conditions rather than relying solely on historical data, facilitating prospective thermal performance assessment across multi-decadal service lives [10]. However, global climate models operate at spatial resolutions that are too coarse for building-scale analysis [21], necessitating downscaling approaches [22–24]. Among available methods, the morphing method [25] remains the most widely adopted statistical downscaling technique [11]. While earlier tools such as CCWorldWeatherGen [26,27], WeatherShift [28], and Meteororm [29] have facilitated this process, they are based on earlier climate model generations and rely on commercial software. Addressing this limitation, Rodrigues et al. [30] developed an open-access Future Weather Generator incorporating the latest shared socio-economic pathway (SSP) scenarios from CMIP6 climate models, which is adopted in the present study. This evolution of weather generation tools reflects the ongoing need to align building performance assessment with current climate projections.

2.1.2. Regional variations in building thermal performance

Generally, the effects of global warming lead to a decrease in heating demand and an increase in cooling requirements [31]. Berardi and Jafarpur [32] reported an average increase of 15–126 % in cooling demand and an average reduction of 18–33 % in heating demand for 16 building prototypes in Canada. However, total energy changes vary significantly among regions. For instance, Jalali et al. [33] revealed a decrease in building energy demand in colder zones and an increase in warmer zones in New Zealand. Similar trends have been observed in the U.S [34]. European studies corroborate these regional patterns across diverse climate zones. Yang et al. [21] conducted a comprehensive assessment of the energy performance of European residential building stocks in 38 cities across five different climate zones, reporting heating decreases of up to 19.7 % in zones 1–2 and cooling demand increases of up to 45 % in zone 5. The implications for thermal comfort have also been considered in future climate change studies. Amaripadath et al. [35] investigated climate change impacts on an nZEB office building in Brussels, reporting an increase in overheating as high as 1.2 °C and a decrease in overcooling as low as 0.3 °C by the end of the century. Rahif et al. [36] further considered discomfort, energy performance and emissions in a temperate climate in Europe, reporting up to 528 % overheating increase by the 2090s. As climate extremes intensify, future peak cooling demand is projected to increase by 25–28.5 % compared to typical conditions [37], while neighborhood peak electricity demand is expected to rise by 4.0 % (near-term), 7.6 % (mid-term), and 16.8 % (long-term) under extreme conditions. These increases pose significant threats to grid capacity and may result in power outages [13]. These findings reveal that the impacts of climate change on building thermal demand are highly region-specific, varying from energy reductions in cold climates to substantial increases in warm zones.

2.1.3. Adaptation and mitigation to climate change

Mitigating the impacts of climate change in buildings requires the implementation of effective energy conservation measures (ECMs) [19]. Building envelopes serve as the primary physical barrier between indoor and outdoor environments, making them critical to achieving energy efficiency. These building envelope components (such as walls and windows) play an important role in regulating heat transfer and represent the most popular ECMs in climate change studies [20]. Traditionally, building envelope optimization has focused primarily on minimizing operational energy consumption. For instance, Pajek et al. [38] evaluated various retrofit measures for energy efficiency in a residential building in Podgorica. They found that thermally insulating opaque envelope components was the most impactful measure, followed by reduced shading setpoints and natural ventilation cooling. Similarly, Rodrigues et al. [17] explored optimal thermal transmittance values for residential buildings in Iran. They demonstrated that ideal future U-values would differ from current optimal values depending on climate conditions. In regions experiencing high cooling loads, buildings require the lowest possible thermal transmittances due to harsh outdoor conditions. This finding aligns with research conducted in tropical Singapore [39], which identified enhanced thermal insulation as a key strategy for optimizing future residential buildings. In a European context, passive design strategies for a single-family house were investigated across five European cities under climate change [40]. The examined parameters included wall and window thermal transmittance, SHGC, shading, and building design variables such as window-to-floor ratio and natural ventilation. Cirrincione et al. [41] investigated the role of green roofs in enhancing energy performance and thermal comfort in urban buildings in Luxembourg and Italy under climate change. They showed that during cooling seasons, green roofs enabled energy savings of 20–50 % for buildings in Esch-sur-Alzette and 2–15 % in Palermo. These studies demonstrate the operational energy benefits of enhanced building envelopes under climate change. However, they consistently overlook the embodied carbon implications associated with the increased use of building materials.

2.2. Life cycle assessment in building design

2.2.1. Embodied and operational trade-off

Focusing on operational performance while neglecting embodied impacts from materials limits the development of whole-life carbon optimization strategies. Notably, the ratio of embodied to operational emissions can approach 1:1 when evaluated over a 50-year reference study period for high energy-efficient buildings [7]. Enhanced thermal performance through additional insulation, advanced glazing and improved assemblies simultaneously reduces operational energy while increasing upfront material emissions. The high proportion underscores the critical importance of addressing the initial upfront carbon emissions alongside subsequent operational impacts from a whole building life cycle assessment (WBLCA) perspective.

For example, Torabi et al. [16] explored over 20,000 designs including geometry, orientation, building envelopes, structural materials, and HVAC system parameters in Canada to identify key drivers of low-carbon buildings during the early design stage. Their analysis showed that electricity grid carbon intensity is the key factor in optimizing carbon footprint. Méndez Echenagucia et al. [42] examined the embodied (cradle-to-gate) and operational carbon of building envelopes across six U.S. cities and identified Window-to-Wall Ratio (WWR) as the most influential design parameter. Bertini et al. [43] evaluated life cycle GHG emissions (LCGHGE) for renovation scenarios in Belgium, reporting that ultra-low energy renovations can reduce the total carbon footprint by 70 % compared to non-renovated buildings, reaching 11.7 kg CO₂eq/(m²·yr). In another study, Gaarder et al. [18] investigated optimal insulation thickness for walls in cold regions under climate change scenarios. Their findings indicate that optimal thickness decreases by 75–100 mm when comparing future projections (2071–2100)

with past conditions (1961–1990) under the RCP8.5 scenario at an emission coefficient of 17 g CO₂eq/kWh. Li and Densley Tingley [44] examined optimal wall insulation thickness for the English housing stock, demonstrating that increasing insulation does not always guarantee improved life cycle performance. Specifically, they found that adding insulation to existing pre-retrofit walls (with thermal transmittances of 0.28 and 0.35 W/m²·K) can actually increase carbon footprint when future climate scenarios are considered. This counterintuitive finding reveals a critical threshold effect: beyond certain thermal performance levels, additional embodied carbon investments yield diminishing or even negative returns under global warming.

2.2.2. Building envelope parameters

Previous research has predominantly focused on adding insulation to external walls [18,44], with limited attention paid to whole building envelope optimization. However, building envelopes comprise integrated systems where windows, walls and roof components collectively serve as the primary barrier against outdoor environmental conditions. The building envelope has impacts on both embodied and operational emissions and thus influences these trade-offs. Saadatian et al. [45] analyzed the embodied impacts of different frame materials and glazing systems. They demonstrated that tempered glass and specialized coatings substantially increase the embodied impacts of window assemblies. These findings highlight the complexity of material selection decisions within envelope optimization frameworks. Architectural design parameters such as WWR also significantly influence the embodied-operational energy balance. This relationship presents a fundamental trade-off: while decreasing window areas or WWR may reduce operational energy consumption by reducing the heat exchange, such modifications may simultaneously result in greater embodied energy requirements due to increased envelope material quantities [46]. Such a dynamic relationship underscores the critical importance of considering whole building envelope parameters for future optimization. Approaches that evaluate individual components, in isolation from broader life cycle considerations, lack the perspective necessary for informed decision-making and may result in suboptimal or counterproductive outcomes.

2.2.3. Hybrid LCA approaches to address truncation errors

The quantification of embodied environmental impacts in building materials requires datasets that capture the full complexity of industrial supply chains and manufacturing processes. However, the development and application of LCA databases present significant methodological challenges that directly influence the accuracy and reliability of assessments. Each database reflects specific regional contexts, employing distinct modeling methodologies and fundamental assumptions that can substantially influence impact calculation outcomes [47]. The magnitude of variability across LCA databases represents a critical challenge for building environmental assessment. Pomponi and Moncaster [5] revealed that the maximum and minimum embodied GHG coefficient for main structural building materials can differ by two orders of magnitude (284 %–1044 %). This extraordinary variation reflects not only regional differences in production technologies and energy systems but also fundamental methodological differences in system boundary definition, allocation methods and impact assessment approaches.

As LCA requires intensive data inputs, dataset availability, transparency and reliability represent essential challenges [48–50]. Currently, the primary approaches in LCA include process-based, input-output, and hybrid approaches that integrate both [51,52]. Process-based LCA represents the most widely adopted approach, utilizing detailed process inventories that trace material and energy flows through specific production pathways [53]. This bottom-up methodology enables precise modeling of foreground processes directly related to building material production but suffers from systematic truncation errors arising from incomplete system boundary definition. These

truncation errors can range from 59–87 % for buildings and products [54], particularly when life cycle inventories cannot capture significant proportions of supply chain inputs [53]. Conversely, input-output LCA methods address these truncation limitations through comprehensive economic modeling that captures complete supply chain interactions within national or regional economies. This top-down approach utilizes economic input-output tables to trace monetary flows through all economic sectors, but may suffer from aggregation errors and temporal mismatches [53,55,56]. Hybrid LCA approaches combine process-based and input-output methods to increase system coverage and enhance the reliability of process analysis [57].

The application of hybrid LCA methodology becomes especially critical for bio-based building materials such as mass timber. Conventional process-based approaches face challenges in accurately representing biogenic carbon dynamics and End-of-Life (EoL) pathways. This is especially relevant for bio-based materials given their dual role as both carbon sinks and potential carbon sources [58]. Given that this research focuses on the relationship between embodied material impacts and operational impacts, rather than detailed processing stage analysis, the research adopts the EPiC database, which utilizes hybrid methodology for the Australian context [57]. Biogenic carbon storage in mass timber elements is not credited in the assessment, in accordance with the 0/0 framework [59,60]. This approach addresses traditional limitations in the process-based method to provide a more robust foundation for optimization studies that balance embodied and operational impacts across WBLCA.

2.3. Research gaps

Contemporary building envelope optimization research demonstrates fundamental methodological limitations that constrain the development of low-carbon design strategies.

First, a substantial body of building envelope optimization research focuses exclusively on operational energy minimization, treating embodied carbon as either a fixed external constraint or omitting it from design objectives entirely, e.g., [61–63]. While this body of work has provided great insights for climate-responsive building design, it does not account for the embodied emissions that become increasingly consequential as electricity grids decarbonize and operational emission savings diminish in relative importance [4,7]. Yet current optimization frameworks do not fully capture the dynamic trade-offs between these competing objectives. Furthermore, existing trade-off studies demonstrate limitations in embodied carbon quantification approaches. Previous studies primarily employ process-based LCA data with narrow system boundaries [15,42,64]. This constrained definition of embodied impacts introduces significant truncation errors by excluding life cycle stages not captured in process-based inventories. These limitations are especially problematic for bio-based materials, where upstream forestry assumptions and downstream EoL pathways substantially influence total embodied carbon. Therefore, this study employs a hybrid LCA methodology that integrates process-based and input-output approaches, providing broader system boundary coverage.

Second, existing embodied-operational trade-off research predominantly relies on static assessment approaches (Table S1), optimizing building envelope design under fixed climate and energy system conditions [16,65]. This temporal disconnect creates a critical knowledge gap: design decisions optimized for present conditions may prove sub-optimal, or even counterproductive, over operational periods exceeding 50 years as climate patterns shift and electricity grids decarbonize. While some studies have incorporated future climate for operational energy assessment and others have explored grid decarbonization scenarios, none simultaneously integrates both temporal dynamics within a unified optimization framework for whole building envelope. The four-scenario framework developed in this study models the temporal evolution of climate change and evolving grids as well as their coupled interactions, advancing substantially beyond the single-condition static

assessments that characterize existing literature.

Third, previous research on embodied-operational trade-offs has predominantly focused on isolated envelope components, particularly external wall insulation thickness optimization [18], while neglecting the interdependencies among complete envelope systems. Windows, walls and roofs interact through complex thermal dynamics. In addition, WWR influences both facade-related material quantities and operational energy consumption. Studies examining insulation optimization in isolation cannot capture how envelope component interactions affect the overall embodied-operational balance, particularly when glazing systems contribute to both embodied carbon and operational energy (through solar heat gain and thermal transmittance). This component-focused approach risks producing optimization recommendations that appear favorable for individual elements but prove sub-optimal when evaluated at the whole-building envelope scale.

To address these interconnected limitations, this research develops an integrated LCA framework that considers climate projections, energy system evolution, and multi-objective building envelope optimization. The framework's contribution lies in its examination of four distinct scenarios:

- 1) Scenario 1 (Static Baseline): establishes performance benchmarks under current conditions
- 2) Scenario 2 (Dynamic Climate): isolates climate change impacts on operational energy
- 3) Scenario 3 (Dynamic Energy): isolates grid decarbonization impacts
- 4) Scenario 4 (Coupled Dynamics): evaluates the full complexity of simultaneous climate and energy transitions

This scenario-based approach enables decomposition of trade-off mechanisms, revealing how each temporal dynamic independently and interactively influences the embodied-operational balance. By introducing Marginal Trade-off Rate (MTR) analysis to quantify efficiency thresholds, the framework provides actionable guidance for climate-responsive design that accounts for both present realities and uncertain futures.

It is worth noting that this research never means to identify the optimal design or achieve the lowest emissions value, as even minor changes to energy-related components would disrupt this equilibrium. This balance remains specific to each case and is influenced by various factors, including but not limited to: local climate, climate models/scenarios, building types, HVAC settings, occupant patterns, reference study periods, energy mix transitions, and the embodied GHG coefficients (types and methods). Instead, this study aims to provide a methodological framework for investigating the intricate relationship between embodied and operational impacts for future research in an era of accelerating climate change. Through integration of climate science, environmental assessment and building simulation optimization, this research contributes to theoretical frameworks for climate-responsive design, the development of sustainable building and policies for the built environment.

3. Materials and methods

The methodological framework integrates four interconnected components to evaluate embodied-operational GHG trade-offs under dynamic climate and energy conditions (Fig. 1). First, a representative 10-storey mass timber office building is established as the case study. Second, future weather scenarios are generated using CMIP6-based morphing tools under SSP245 and SSP585 pathways, providing climate inputs for three temporal periods. Third, a WBLCA framework is developed using the EPiC hybrid LCA database to quantify embodied GHG emissions, with operational emissions computed through four distinct scenario formulations capturing static, climate-dynamic, grid-dynamic, and fully coupled conditions. Fourth, multi-objective optimization is implemented using the NSGA-II algorithm to explore the Pareto

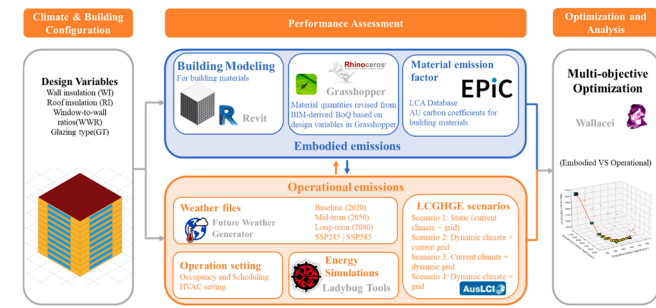


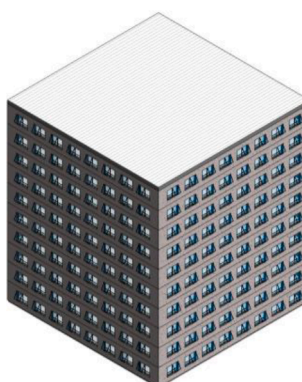
Fig. 1. Methodological framework for embodied-operational emissions trade-off analysis.

frontier of embodied-operational trade-offs across all scenarios and climate zones.

3.1. Building geometry and configuration

The case study employs a representative 10-storey commercial office building in Australia, which has been widely investigated in previous case studies [66–68]. Table 1 presents the complete building specifications, including geometric parameters, occupancy characteristics and HVAC system configurations. Recognizing Australia's rapid construction development and urgent emission reduction requirements, the structural system employs mass timber as an increasingly prevalent sustainable solution in Australia [69–71]. In addition, its lower embodied carbon relative to conventional concrete and steel [72,73], makes it a particularly instructive test case for embodied-operational trade-off

Table 1
Building geometry and configuration parameters for life cycle assessment.

Parameter	Specification
Visualization	
Typology	Office
Gross floor area (m ²)	9985.6
Footprint dimensions (m)	31.6 × 31.6
Floor-to-floor height (m)	3.6
Window-to-wall ratio (baseline)	30 %
Analysis scope	Superstructure (excluding foundations)
Frame system	CLT shear walls with glulam post-and-beam framing
Occupant density (Person/m ²)	1/10
Equipment load (W/m ²)	11
Lighting load (W/m ²)	4.5
Building operations schedule (HVAC)	National Australian Built Environment Rating System (NABERS) [77] VAV with PFP Boxes
Heating setpoint(°C)	20
Cooling setpoint(°C)	24

analysis. The primary structure employs glulam for beams and columns alongside cross-laminated timber (CLT) for walls and roofs, reflecting contemporary sustainable construction practices. The detailed design of mass timber construction references the CLT handbook [74], Dataholz database [75], and Australian study [76] to reflect common industry practices. The wall assembly incorporates a multi-layered system comprising: 25 mm timber cladding, vertical battens, variable insulation thickness as a key parametric variable, 160 mm CLT, and 13 mm interior gypsum board finish. Similarly, the roof assembly consists of 50 mm gravel, waterproof sealing sheet, variable insulation thickness, an additional sealing membrane, 160 mm CLT, 120 mm cavity fill, and 13 mm plasterboard ceiling finish.

3.2. Future weather scenarios and building simulations

The dynamic nature of climate change generates fluctuating environmental parameters that fundamentally challenge the optimization balance while simultaneously reducing building resilience to evolving climatic conditions. This study utilized *Future Weather Generator* [30], which features an open access morphing weather file tool based on CMIP6 climate models. Two representative scenarios were selected to capture the range of potential climate futures: SSP245 representing moderate warming scenarios, and SSP585 reflecting high intensity warming trajectories. These scenarios represent plausible climate futures, ranging from moderate mitigation success to continued high emissions, thereby enabling assessment of building performance under both optimistic and pessimistic climate trajectories. The climate impact assessment encompasses three temporal periods: current baseline conditions, mid-term projections (2050), and long-term scenarios (2080). Building energy modeling was conducted using Grasshopper for Rhino coupled with Ladybug Tools (EnergyPlus engine) for hourly time-step simulations. The building is modelled as fully electric, with no gas-based energy systems considered. Operational GHG emissions are calculated from electricity consumption across all simulated end uses, including heating, cooling, ventilation, lighting, and equipment loads.

3.3. Life cycle assessment methodology

The assessment integrates both embodied and operational impacts to establish a complete WBLCA over a 60-year reference study period. Embodied GHG emissions are quantified using the EPiC database, which employs a hybrid life cycle inventory methodology for Australian construction materials [57]. This approach addresses truncation errors inherent in conventional process-based LCA while providing regionally appropriate environmental impact coefficients that reflect local supply chain characteristics and energy systems [78,79]. Material quantity assessment derives from the parametric building model integrated with detailed bills of quantities extracted from Revit. The embodied assessment focuses on envelope-related materials within the superstructure (the above-ground structural system excluding foundations), including insulation systems, structural timber components, and window assemblies (see Supplementary Material for detailed material scope). Window system specifications and embodied impact data are adapted from established research [45,80], with regional adjustments applied using EPiC database coefficients to ensure alignment with Australian construction contexts. Details about embodied impact quantification and operational energy simulation are based on previous studies [46,58].

Total embodied GHG emissions (E_{GHG}) are calculated using the following Equation:

$$E_{GHG} = \sum (Q_i \times EF_i)$$

Where:

E_{GHG} = Total embodied GHG emissions (kg CO₂eq/m²)

Q_i = Quantity of material i (kg or m³)

EF_i = Embodied emission factor for material i (kg CO₂eq per unit)

Operational GHG emissions are determined by coupling annual energy consumption with electricity grid emission factors, with distinct computational formulations applied across four analytical scenarios to capture varying temporal and environmental dynamics.

For Scenario 1 (Static baseline),

$$O_{GHG} = EC_{2020} \times EF_{2020} \times RSP$$

For Scenario 2 (Dynamic climate change),

$$O_{GHG} = EF_{2020} \times \int_{2020}^{2080} EC(t)dt$$

For Scenario 3 (Dynamic energy system evolution),

$$O_{GHG} = EC_{2020} \times \int_{2020}^{2080} EF(t)dt$$

For Scenario 4 (Dynamic climate and energy),

$$O_{GHG} = \int_{2020}^{2050} EC(t) \times EF(t)dt + \int_{2050}^{2080} EC(t) \times EF(t)dt$$

Where:

O_{GHG} = Total operational GHG (kg CO₂eq/m²)

EC_{2020} = Current energy consumption at baseline weather (kWh/m²)

EF_{2020} = Current electricity emission factor (kg CO₂eq/kWh)

$EC(t)$ = Climate-adjusted annual energy consumption at time t (kWh/m²)

$EF(t)$ = Time-varying emission factor (kg CO₂eq/kWh)

RSP = Reference study period (60 years)

Static baseline conditions (S1) maintain constant Australian electricity emission factors and building energy performance throughout the analysis period. The S2 scenario preserves current emission factors while incorporating climate-induced variations in energy consumption patterns. The S3 scenario maintains current climatic conditions while implementing temporally evolving emission factors reflecting projected electricity grid decarbonization trajectories. The S4 scenario integrates both climate-induced energy consumption changes and temporally varying emission factors. The study assumes linear progression in electricity emission factors and building energy consumption across two periods: 2020–2050 and 2050–2080. Electricity mix emission factors for future periods (2050 and 2080) are derived from the AusLCI database (See Supplementary Material Table S2). Importantly, the specific mechanisms driving emission factor transitions extend beyond this study's analytical scope, as the research emphasizes building envelope design optimization rather than energy system transition modeling. It should also be noted that this study employs annual average electricity emission factors for each state and territory, rather than dynamic (hourly or sub-hourly) grid emission factors that reflect temporal fluctuations in electricity generation mix [43,81,82]. Analyzing dynamic grid emission factors falls outside the scope of this study due to limited data availability. In addition, incorporating sub-hourly temporal resolution would substantially increase computational complexity.

The Marginal Trade-off Rate (MTR) quantifies the rate at which embodied emissions change relative to operational emission variations within building envelope optimization frameworks. The MTR is mathematically expressed as:

$$MTR = \frac{\partial E_{GHG}}{\partial O_{GHG}}$$

Where:

- E_{GHG} = Embodied GHG emissions changes (kg CO₂eq/m²)
- O_{GHG} = Operational GHG emissions changes (kg CO₂eq/m²)

3.4. Parametric design variables

The parametric analysis varies key envelope design parameters (Table 2) based on previous building envelope studies [42,83–85]. Wall insulation thickness varies from 0.02 to 0.40 m with roof insulation ranging from 0.03 to 0.60 m, reflecting 0.455–0.085 W/m²·K (walls) and 0.366–0.059 W/m²·K (roofs) [18]. WWR spans from 0.2 to 0.7, representing the full range of architectural typologies from conventional office buildings to contemporary high-glazing designs [42,84]. Window system specifications encompass thermal transmittance values ranging from 0.74 to 5.84 W/m²·K and solar heat gain coefficients from 0.33 to 0.88, reflecting the diversity of glazing technologies and frame materials available in current markets [45,80].

3.5. Optimization analysis

The optimization framework employs the Wallacei X plugin for Grasshopper, implementing the Non-dominated Sorting Genetic Algorithm II (NSGA-II) for multi-objective optimization [86]. NSGA-II was selected over alternative algorithms such as Multi-Objective Particle Swarm Optimization (MOPSO); the non-domination sorting and crowding distance mechanisms inherent to NSGA-II produce well-distributed Pareto fronts [87,88] that facilitate visual and analytical interpretation of trade-off relationships, a primary objective of this study. NSGA-II has been widely used in previous building performance optimization studies [89–91] and is natively implemented in the Wallacei X plugin for Grasshopper [86], enabling tight integration with the EnergyPlus simulation environment without inter-software data exchange overhead. All simulations were conducted on a desktop computer equipped with an Intel i9–12,900 processor, utilizing a population size of 30 individuals across 30 generations. The workflow focuses on investigating the trade-off relationship between embodied and operational emissions rather than pursuing a singular optimal solution that minimizes their combined total. The multi-objective framework facilitates systematic exploration of the Pareto frontier, revealing how envelope design decisions influence the balance between upfront material emissions and long-term operational performance. By treating embodied and operational GHG as competing objectives, the optimization algorithm explores diverse design configurations that represent different points along the trade-off spectrum. Such a method prevents convergence to a single solution and enables comprehensive analysis of the embodied-operational relationship.

4. Results and discussion

This section presents an integrated analysis of embodied-operational GHG emissions trade-offs, examining how climate change and energy system evolution reshape the trade-off relationships and building envelope design strategies. The analysis progresses from baseline performance through dynamic climate and energy scenarios. A summary of

Table 2
Building envelope parameters variables.

Parameter	Units	Range/Values	Increment
Wall insulation thickness	m	0.02–0.40 (0.455–0.085 W/m ² ·K)	0.02
Roof insulation thickness	m	0.03–0.60 (0.366–0.059 W/m ² ·K)	0.03
Window-to-wall ratio	/	0.2–0.7	0.1
Windows	/	Aluminum, PVC, Wood 24 type (8 per frame material)	/
Window U-value	W/m ² ·K	0.74–5.84	/
Window SHGC	-	0.33–0.88	/

Note: Insulation thickness ranges correspond to U-values of 0.455–0.085 W/m²·K (walls) and 0.366–0.059 W/m²·K (roofs).

Pareto frontier optimization results for embodied, operational, and LCGHGE ($\text{kg CO}_2\text{eq/m}^2$) across all cities and scenarios is provided in Table S3 in the Supplementary Material.

4.1. Baseline scenario performance and trade-off quantification

The multi-objective optimization analysis under current conditions already reveals profound spatial heterogeneity in embodied-operational carbon trade-offs across Australian cities (see Figs. S1-S9 in SM). The x-axis represents embodied GHG emissions ($\text{kg CO}_2\text{eq/m}^2$), the y-axis represents operational GHG emissions, and the z-axis represents life cycle GHG emissions. Each point on the frontier represents a distinct building envelope design configuration generated through multi-objective optimization. The optimization results show minimum LCGHGE ranging from 1426 in Hobart to 8322 $\text{kg CO}_2\text{eq/m}^2$ in tropical Townsville (Fig. 2), representing a six-fold variation. This disparity stems primarily from differences in operational emissions associated with heating and cooling energy consumption driven by local weather parameters, as well as the varying electricity emission factors among these cities. In addition, comparison with other studies should be with caution, as the schedules, occupant behavior and building size vary greatly in different building types. Cities in intermediate climate zones demonstrate the lowest LCGHGE values that reflect their distinct energy consumption patterns: Melbourne (5993 $\text{kg CO}_2\text{eq/m}^2$), Perth (4451),

and Sydney (5434) each exhibit emissions profiles that correspond to their specific climatic conditions and energy mix.

The Pareto frontier characteristics reveal electricity-dependent optimization effectiveness. Hobart demonstrates the most pronounced U-shaped Pareto front morphology, with total emissions ranging tightly between 1426–1535 $\text{kg CO}_2\text{eq/m}^2$. This pattern directly correlates with Tasmania's exceptionally low electricity emission factor of 0.16 $\text{kg CO}_2\text{eq/kWh}$, which fundamentally alters the optimization pattern. This is consistent with previous Canadian studies which report relatively low electricity emissions: 1.3, 29 and 12.3 $\text{g CO}_2\text{eq/kWh}$ for Quebec, Toronto and Vancouver [92], and 36.3 $\text{g CO}_2\text{eq/kWh}$ for Toronto [32]. Under such conditions, operational energy consumption changes produce marginal emissions impacts. However, the ongoing electrification of heating and transportation sectors introduces additional complexity, potentially increasing electricity demand. This regional variation in electricity carbon intensity creates fundamental trade-offs between energy source selection and emission intensity optimization for heating and cooling. Such complexity extends beyond the current study's analytical scope but represents an area for future investigation.

The local condition, including climate and energy mix, determines not only absolute performance levels but also optimization effectiveness, with total emissions variability ranging from very limited in cool temperate (Hobart: 109 $\text{kg CO}_2\text{eq/m}^2$ optimization range: maximum minus minimum Pareto frontier emissions) to substantial in tropical

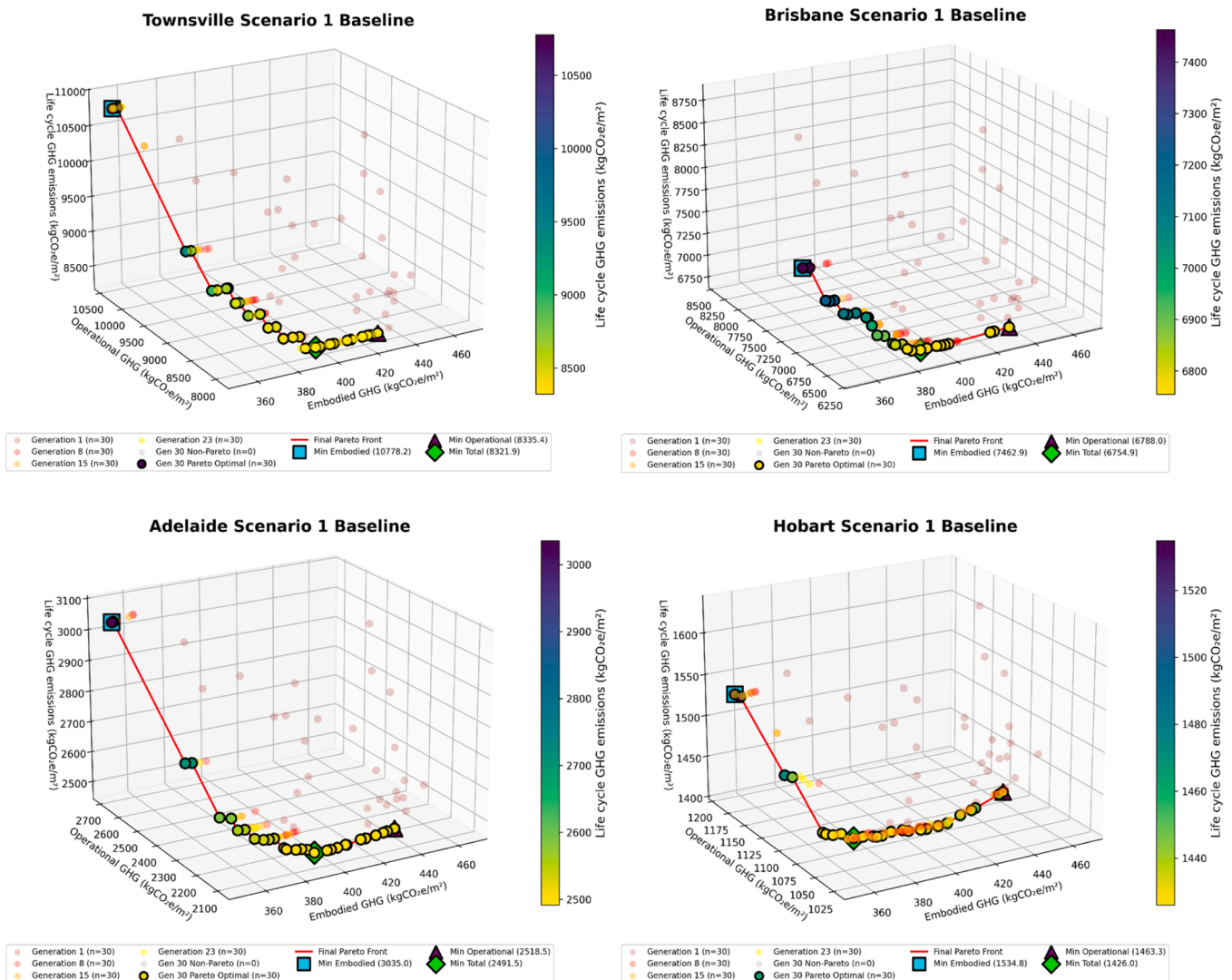


Fig. 2. Multi-generation Pareto frontier evolution under the static baseline scenario (S1) for Townsville, Brisbane, Adelaide and Hobart.

conditions (Townsville: 2457 kg CO₂eq/m² optimization range). Moderate climate zones exhibit intermediate variability, with Adelaide (544 kg CO₂eq/m²), Perth (1199), Melbourne (1057), and Sydney (1515) showing a clear association with climatic severity. This pattern suggests that tropical, high-emission contexts offer the greatest potential for envelope-driven performance improvements. These findings provide a spatially differentiated evidence base that may inform the prioritization of regulatory attention and policy efforts, though direct evaluation of policy instruments falls outside the scope of this study.

Low emission factors also create a dilemma. In Hobart, the lowest LCGHGE occurs with WWR of 0.6, contrasting sharply with the optimal WWR of 0.2 observed in other cities. This counterintuitive result reflects the interaction of two mechanisms. First, in Tasmania's mild heating-dominated climate with an exceptionally low electricity emission factor (0.16 to 0.142 to 0.134 kg CO₂eq/kWh), the marginal operational penalty associated with increased glazing area, through greater thermal transmittance and variable solar heat gain, is negligible in emissions terms, even though the absolute energy consumption changes. Second, a higher WWR reduces the total quantity of opaque wall assembly materials (cladding, CLT panels, insulation layers), lowering embodied GHG. This embodied material saving then dominates the total life cycle calculation changes because the operational penalty is suppressed by the low emission factor. In cities with higher electricity emission factors, the same increase in WWR produces a far larger operational emissions penalty, amplified multiplicatively by the emission factor, which outweighs the embodied savings and pushes optimal solutions toward lower WWR values.

This numerically driven design proves impractical for several important reasons. First, energy use is not without cost, as high energy consumption increases occupant expenditure. Average annual emission factors also mask real emissions variability, as hourly grid emission dynamics (marginal emissions at hourly resolution) are not captured by annual averages [81]. Second, low-emission and energy-efficient technologies may trigger rebound effects where occupants consume more energy than predicted [93]. Current modeling assumptions exclude dynamic occupant behavior patterns, which significantly affect actual energy consumption. Third, maximizing energy consumption stresses grid infrastructure and increases power outage risk, especially during more frequent and severe extreme weather events [13]. It should also be noted that while all optimal solutions favor wood windows due to their lower emissions, in practice, factors beyond emission coefficients matter, including cost, maintenance, durability, air infiltration, and thermal bridge effects. Those details are difficult to capture in emission-focused optimization and research.

The Pareto frontier results also reveal a significant asymmetry in the variation ranges between embodied and operational emissions. Across all climate zones, the embodied GHG emissions of Pareto-optimal solutions demonstrate remarkably constrained optimization ranges (maximum - minimum), typically spanning 60–100 kg CO₂eq/m² (19.20–25.70 % relative variation), while operational emissions exhibit substantially broader ranges from 150–2500 kg CO₂eq/m² (11.90–38.30 % relative variation). Specifically, Townsville shows embodied emissions varying by about 71.74 kg CO₂eq/m² (353–424 kg CO₂eq/m²) compared to operational emissions spanning 2514.61 kg CO₂eq/m² (7911–10,426 kg CO₂eq/m²), representing a 35:1 ratio in absolute variability.

Such a narrow embodied GHG range occurs because structural material selections remain constant, and embodied GHG is driven by material quantities alone. This is echoed by Torabi et al. [16], which demonstrated that structural materials are the most influential parameters affecting total emissions in low grid-carbon cities. In contrast, the substantial operational emission ranges reflect the cumulative nature of long-term energy consumption in a large commercial office building over a 60-year study period, where envelope-driven differences in thermal performance compound significantly over time. This imbalance is further amplified by location-dependent factors: cities with high

electricity emission intensities and thermally demanding climates, such as tropical Townsville, exhibit particularly wide operational emission ranges, as even moderate changes in envelope performance translate into large cumulative GHG differences when multiplied by high emission factors over six decades. Embodied GHG, by contrast, remains comparably constrained across locations because the coefficients are fixed and independent of climate or energy mix. This suggests that operational optimization offers significantly greater leverage for total emissions reduction than embodied GHG management within the current analytical scope, particularly in high carbon intensity and energy-intensive contexts.

Across all baseline Pareto solutions, embodied GHG accounts for 3.3 %–29.5 % of total emissions. These proportions reflect the large variation in operational emissions resulting from climatic factors and energy intensity of local electricity grids. This aligns reasonably with Robati et al. [70] who report an embodied proportion of approximately 27 % under a static electricity grid assumption based on an 18-storey building in Sydney. The MTR quantifies the rate at which embodied carbon increases relative to operational carbon reductions. A key observation across all cities is the existence of a trade-off efficiency threshold beyond which additional embodied carbon investments yield progressively smaller operational benefits. This threshold occurs after achieving the lowest LCGHGE (Fig. 3). Beyond that point, most design combinations exhibit unfavorable trade-offs (MTR > 1), highlighting the optimization area where meaningful efficiency gains occur.

Reducing energy consumption from poor to moderate efficiency levels (e.g., halving initial consumption) proves relatively straightforward through conventional envelope improvements, as demonstrated by steep performance gains in early optimization stages. However, pursuing further optimization to achieve marginal energy reductions incurs disproportionately high embodied GHG and economic costs, raising fundamental questions about resource allocation efficiency and cost-effectiveness thresholds. The law of diminishing returns manifests as MTR values escalate rapidly beyond optimal insulation thicknesses, where additional material investments fail to justify their embodied GHG burden. Specifically, over-insulation yields diminishing marginal improvements compared to applying insulation to poorly insulated buildings, reflecting the non-linear relationship between thermal resistance and insulation thickness.

Northern European research on optimal insulation thickness in cold climate case studies found that when electricity emission factors are low (9–17 g CO₂eq/kWh), there is a U-shaped relationship between insulation thickness and total emissions [18]. When insulation thickness approached the optimum value, changes in LCGHGE were marginal due to the nonlinear relationship between U-value and insulation thickness.

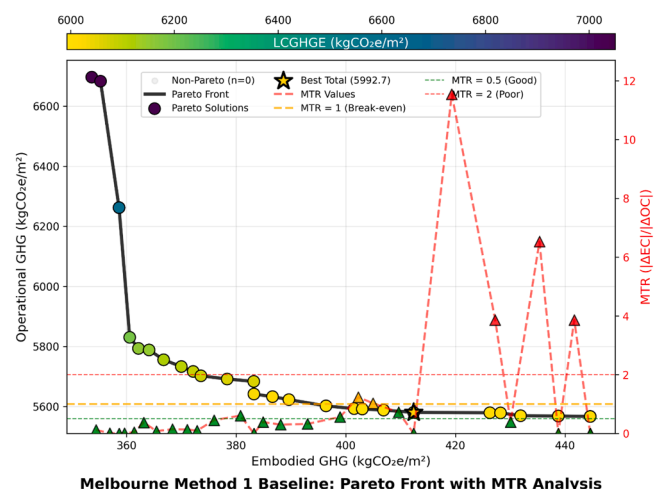


Fig. 3. Pareto front with MTR analysis for Melbourne S1.

When electricity emission factors are higher (132–207 g CO₂eq/kWh), the optimal thickness exceeded the parameter range, but a reduced impact trend was observed. The trend is also reflected in the current study through the sharp increase in MTR after optimal design, as well as the marginal impact on total LCGHGE. Except in Hobart, the optimal design for lowest emissions occurs early in the Pareto front. This demonstrates that electricity emission factors have a significant impact on optimization results by directly influencing operational emissions trends.

The MTR analysis reveals that traditional operational energy minimization approaches may inadvertently increase LCGHGE when pursued beyond optimal thresholds. The pursuit of zero operational energy without considering embodied trade-offs can lead to carbon-intensive building solutions that contradict climate mitigation objectives. These findings challenge conventional building rating systems that prioritize

nearly zero or zero operational energy. The research demonstrates that optimal building performance occurs at intermediate levels of thermal enhancement, not at the extremes of either minimal or maximal envelope optimization. Poorly performing conventional buildings may derive substantially greater benefits from envelope renovation, as evidenced by the steep improvement gradients observed at the initial segments of the Pareto frontiers. This finding underscores the importance of targeting retrofit interventions toward building stocks with the greatest potential for performance enhancement first, rather than pursuing zero energy consumption targets in high-performing buildings.

Such observations extend beyond building-level optimization to broader climate policy design. The spatial and threshold-based patterns identified here suggest that demand-side interventions may yield differentiated returns across contexts, an observation that could inform the design of carbon pricing or emissions trading mechanisms [94]. This

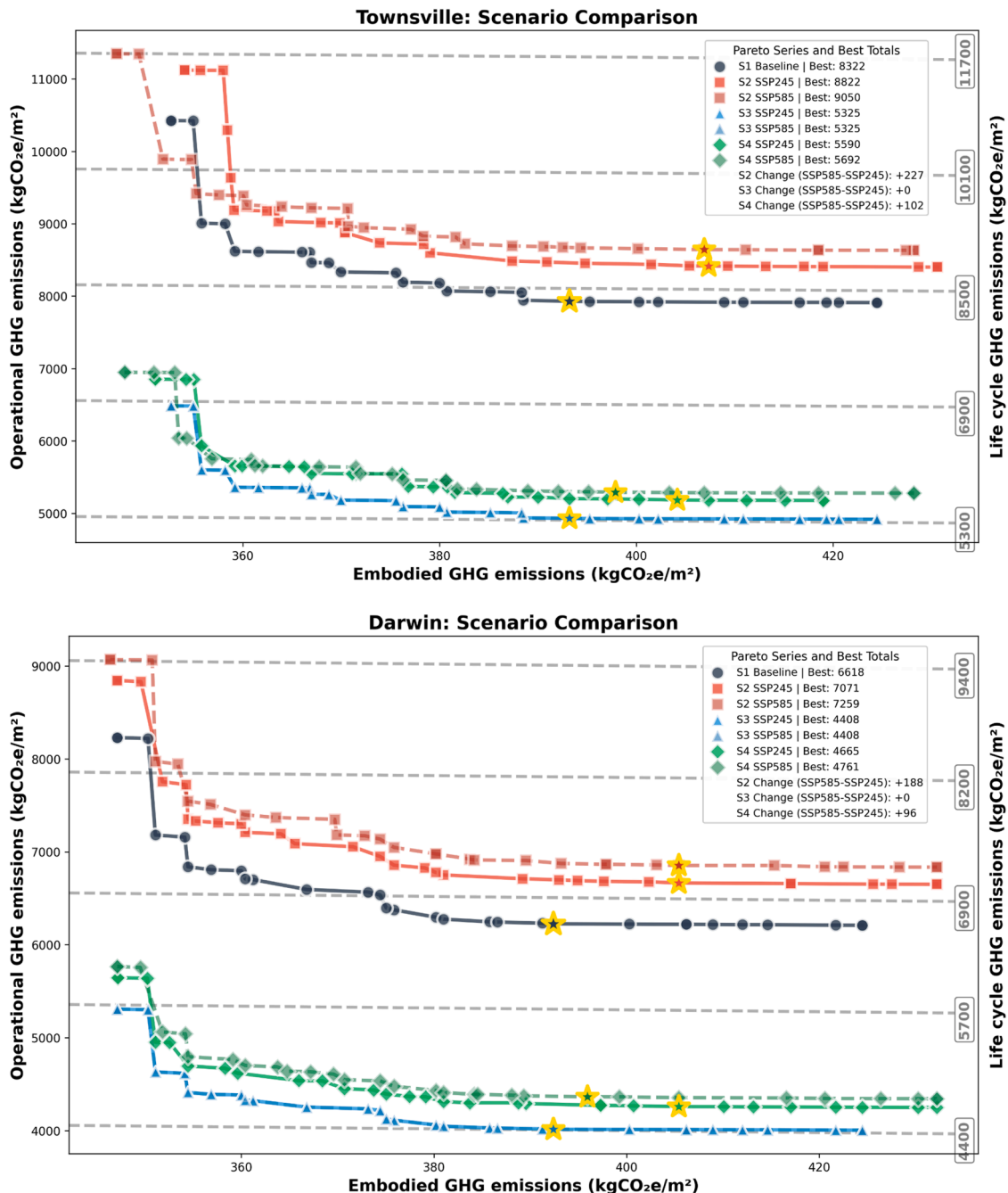


Fig. 4. Life cycle optimization comparison for Townsville and Darwin.

occurs because pursuing marginal efficiency gains in high-performance buildings leads to disproportionately higher costs relative to the associated carbon savings. Understanding the embodied-operational trade-off curves across different building typologies and regions could theoretically support more efficient deployment of limited decarbonization capital. However, policy instruments introduce fundamental dimensions of climate equity that extend beyond technical optimization to encompass distributional justice concerns. This would require integration with economic, climate science and social equity considerations.

4.2. Dynamic scenarios on life cycle impacts

Building emissions are inherently dynamic. The dynamic scenario framework addresses three distinct future pathways: climate change impacts (S2), energy system decarbonization (S3), and coupled climate-energy dynamics incorporating both factors simultaneously (S4).

4.2.1. Climate change impacts on embodied-operational balance

Climate change impact magnitudes vary substantially across Australian climate zones, with tropical regions exhibiting the greatest sensitivity to warming scenarios. A predominant pattern emerges across Australian cities where S2 Pareto frontiers are positioned consistently above S1 baselines, indicating that climate change impacts increase total LCGHGE regardless of envelope design optimization. Climate change imposes performance penalties of 2.46 %–6.84 % in LCGHGE across Australian cities, with notable exceptions in heating-dominant regions like Hobart and Canberra (–0.1 % to 0.01 %) under SSP245 scenarios. Under the most severe scenarios, optimal LCGHGE deteriorates by 3.16 %–9.69 % compared to baseline conditions (exclude Hobart and Canberra).

Tropical and subtropical regions exhibit the most pronounced performance degradation (Fig. 4), with Pareto frontier displacement ranging up to 1000 kg CO₂eq/m² across optimization ranges. In Townsville, the minimum LCGHGE increases from 8321.88 kg CO₂eq/m² (S1) to 8822.46 and 9049.70 kg CO₂eq/m² (S2) under SSP245 and SSP585, respectively, representing a 6.02 %–8.75 % deterioration in optimal performance. This increase reflects intensified cooling demands that outweigh marginal heating reductions, with operational emissions rising from 7928.69 to 8415.13 and 8642.82 kg CO₂eq/m² respectively while embodied emissions remain relatively stable.

Similarly, Darwin demonstrates significant climate sensitivity with minimum LCGHGE increasing from 6618.07 to 7071.06 kg CO₂eq/m² (6.84 % increase) under SSP245 and 7259.09 kg CO₂eq/m² under SSP585 (9.69 % increase). The operational component rises from 6225.68 to 6665.67–6853.70 kg CO₂eq/m², indicating that cooling-dominated climates face substantial performance penalties under warming scenarios.

This consistent degradation pattern indicates that envelope optimization alone may not compensate for climate-driven operational energy increases in tropical regions. These findings align with studies conducted in Taiwan, which reported that no individual building envelope strategies, including improvements to U-values of walls and roofs, window properties (U-value and SHGC), or shading systems, could maintain future cooling demands at current levels in residential buildings [95]. This raises fundamental questions about the adequacy of passive building strategies under warming scenarios [20,96]. Research by D'Agostino et al. [96] demonstrates that the effectiveness of nighttime ventilation diminishes as climate change increases summer nighttime temperatures, thus emphasizing the need for active cooling measures in a warming climate. However, these limitations may be context-dependent based on regional climate characteristics. In tropical regions, mitigating cooling demand becomes increasingly challenging as temperature increases can trigger non-linear surges in cooling requirements.

Temperate cities demonstrate variable climate responses, with most showing moderate to marginal Pareto frontier elevation despite

expectations of improved heating-cooling balance. Perth (Fig. 5) exhibits minimal degradation under SSP245 but shows more pronounced upward displacement under SSP585. Melbourne shows consistent performance degradation across both SSP scenarios with ~200 kg CO₂eq/m² increase in LCGHGE. Adelaide shows minimal change, with total LCGHGE remaining essentially stable (2491–2570 kg CO₂eq/m²). This pattern reflects complex heating-cooling dynamics, where warming reduces winter energy demands while increasing summer requirements. Sydney exhibits similar stability under SSP conditions (5434 vs 5769–5870 kg CO₂eq/m²). These findings challenge assumptions that temperate climates inherently provide resilience to climate change impacts.

Canberra and Hobart (Fig. 6) represent notable exceptions where S2 and S1 Pareto frontiers exhibit convergence rather than displacement even under the most severe climate scenarios. Hobart demonstrates remarkable frontier stability with minimal vertical separation. This indicates that heating load reductions in cool climates can indeed offset cooling load increases in the optimization under specific climatic and building energy system conditions. In Hobart, the minimum LCGHGE shows minimal variation across climate projections (1421–1426 kg CO₂eq/m²), with operational emissions actually decreasing slightly from 1059 to 1046–1048 kg CO₂eq/m². However, this apparent climate resilience requires careful examination of underlying assumptions. The minimal climate change impact may reflect Tasmania's unique combination of maritime moderation and low electricity emissions factors.

Importantly, these findings focus on annual energy totals rather than peak load implications and thermal comfort boundary effects, which may reveal more severe impacts. The upward displacement of optimization frontiers suggests that climate change fundamentally alters the physics of building thermal performance in ways that envelope optimization cannot fully address. Passive building strategies may have reached effectiveness limits under warming scenarios. Mechanical system efficiency improvements, renewable energy integration, and demand management strategies may be more effective climate adaptation measures. Active carbon optimization strategies can be coupled with the passive envelope framework developed here. Battery energy storage systems (BESS) represent a particularly promising integration, enabling buildings to decouple electricity consumption from periods of high grid carbon intensity through strategic load shifting and demand response. Increasing renewable energy penetration introduces supply intermittency. BESS-enabled demand flexibility could reduce the marginal emission intensity of building operational consumption beyond what passive envelope optimization alone can achieve. Recent studies demonstrate the practical significance of such integration. Amaripadath et al. [82] investigated plug-in BESS for net zero energy buildings across G7 countries, reporting that energy storage systems substantially reduce nighttime grid electricity consumption by shifting demand away from carbon-intensive periods, though charging and discharging losses introduce efficiency penalties. Notably, carbon-free electricity utilization increased by up to 32.3 % in Tokyo, highlighting the critical dependence of BESS effectiveness on local grid carbon intensity. Similarly, Li et al. [97] demonstrated that optimized BESS deployment across 72 buildings achieved significant cost reductions (C\$ 311 K) and emission savings of 471 tonnes CO₂eq over three winter months. Beyond standalone BESS applications, systems combining BESS with thermal energy storage, PV systems and heat pumps for optimal sizing [98], as well as electric vehicle integration within building microgrids [99], further expand the demand-side flexibility potential. Passive strategies reduce the baseline energy demand that active systems must manage, while BESS and demand response technologies optimize the timing and carbon intensity of remaining consumption. Such integrated approaches are particularly relevant for the tropical and subtropical cities identified in this study, where the largest gaps exist between passive envelope optimization potential and climate-driven operational penalties that envelope improvements alone cannot fully address.

This strategic reorientation toward active systems becomes

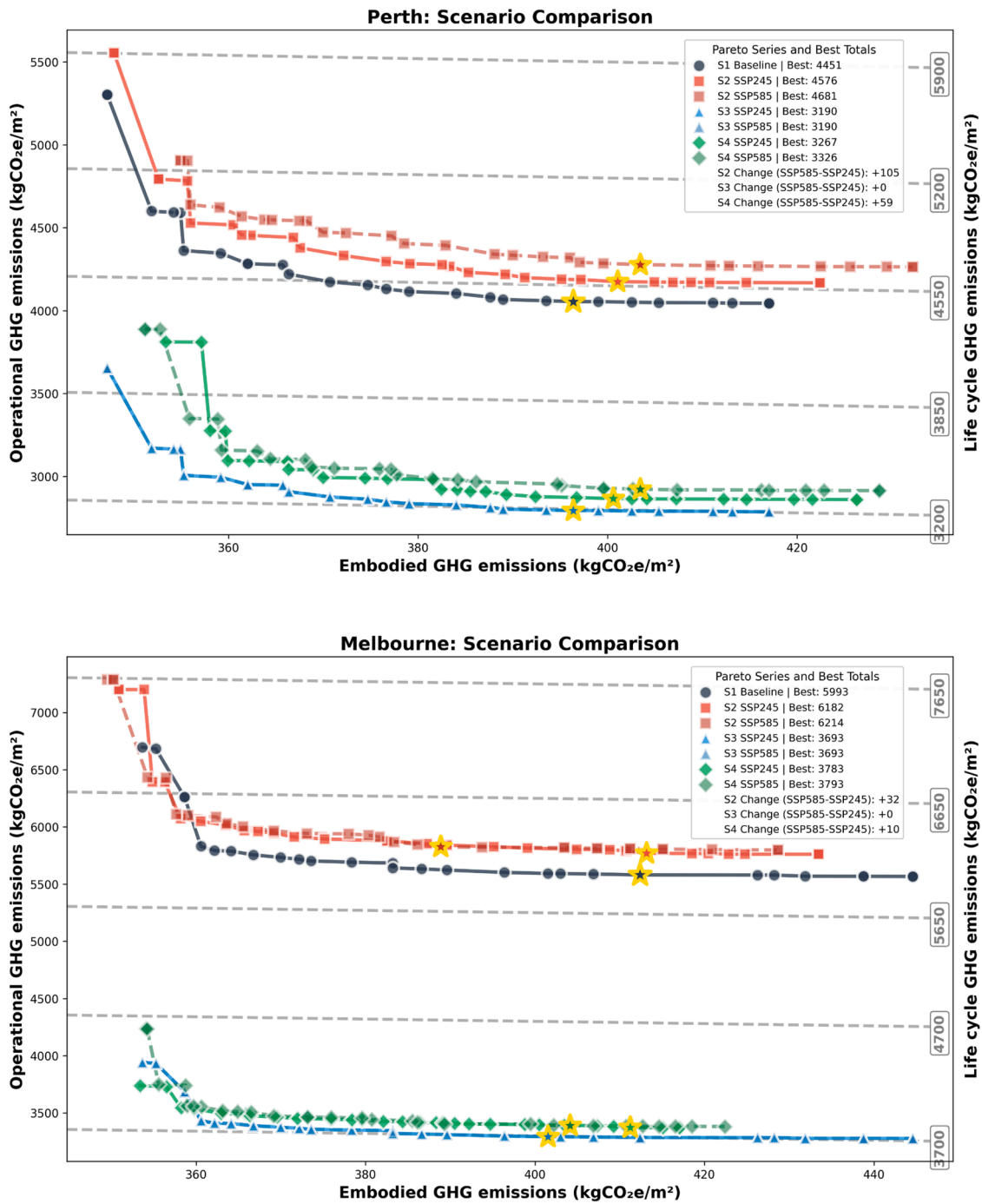


Fig. 5. Life cycle optimization comparison for Perth and Melbourne.

especially critical as climate projections indicate intensifying extreme events. Beyond building envelope considerations, operational energy performance is intrinsically linked to HVAC system design and efficiency. Gauch et al. [100] demonstrated that mechanical ventilation with heat recovery represents one of the most effective interventions to reduce operational energy consumption, especially in hot, humid and cold climate office buildings. However, HVAC and other plumbing systems also contribute to embodied impacts through components such as ductwork, piping, and mechanical equipment, which are frequently overlooked in building-level LCA due to data availability constraints and collection challenges [101]. This warrants comprehensive investigation in future WBLCA, particularly as climate adaptation strategies increasingly emphasize mechanical system performance.

The regional variation in climate impact magnitude highlights that building performance is highly location-specific, with cities showing the largest frontier displacement requiring different adaptation strategies than those exhibiting convergence of the embodied and operational trade-offs.

4.2.2. Energy system evolution impacts on the embodied-operational balance

The Pareto frontier comparisons demonstrate vertical stratification between current grid conditions (S1&2) positioned in upper emission territories and decarbonized scenarios (S3&4) concentrated in lower performance zones, with the magnitude of separation directly correlating with baseline electricity emission intensity. Notable exceptions

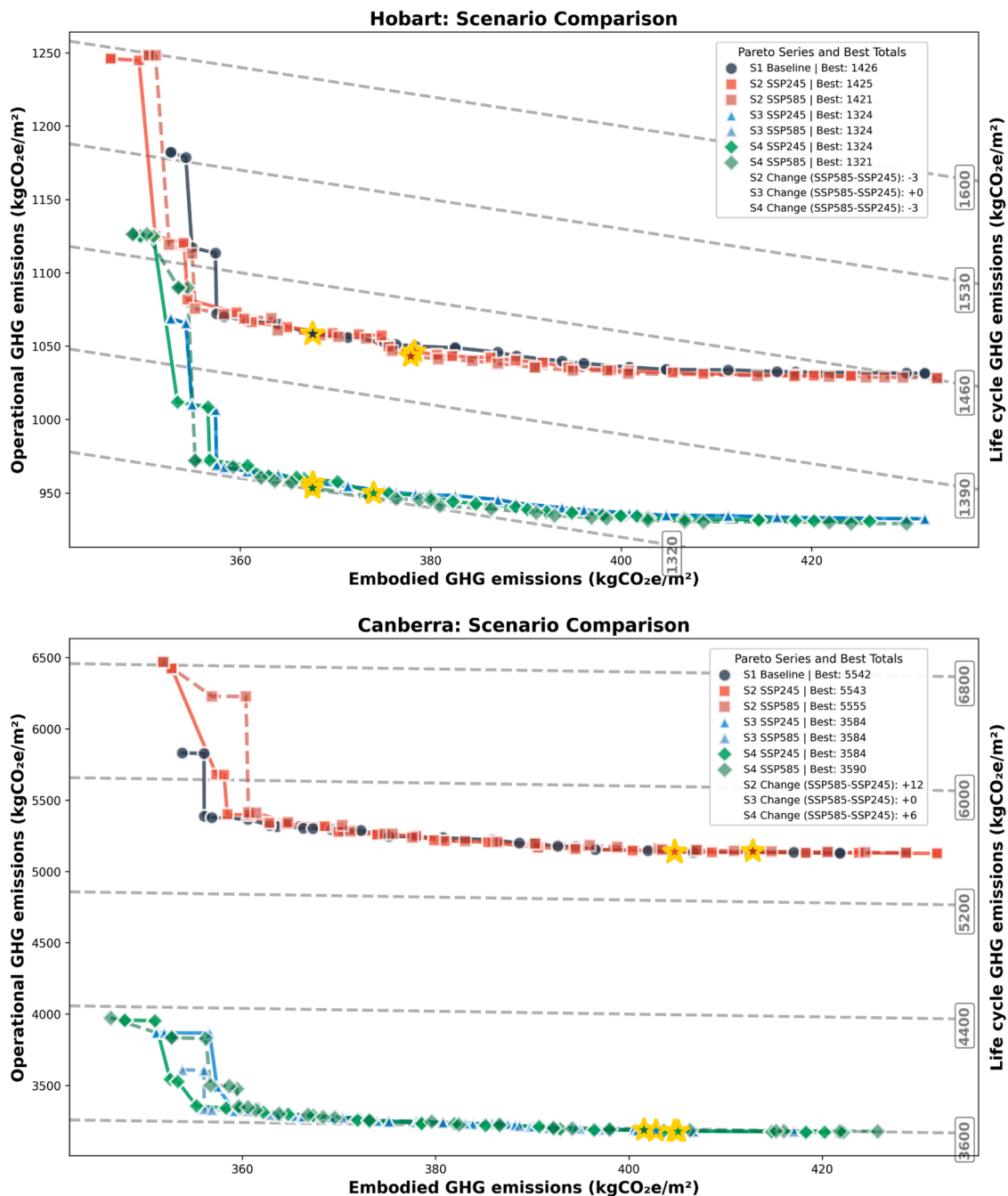


Fig. 6. Life cycle optimization comparison for Hobart and Canberra.

include Hobart and Adelaide, which exhibit minimal changes in electricity emission intensity, resulting in clustered rather than vertically stratified results. Energy system evolution scenarios demonstrate substantial LCGHE improvements, representing 28.32–38.37 % reductions in high emission factor regions, moderate improvements in Hobart (7.12 % reduction), but a 7.59 % increase in Adelaide.

Cities with high current emission factors, such as Brisbane and Townsville, exhibit pronounced vertical separation between current grid (S1&2) positioned in upper performance regions and decarbonized scenarios (S3&4) concentrated in lower emission zones. This stratification is most pronounced in coal-dependent grids where the vertical displacement between scenario clusters exceeds 2000–3000 kg CO₂eq/m² across the entire optimization frontier. Specifically, Townsville's 36.0 % emission reduction (8322 to 5325 kg CO₂eq/m²) and Brisbane's 35.6 % improvement (6755 to 4347 kg CO₂eq/m²) demonstrate

complete optimization restructuring where embodied carbon share increases from ~5 % to ~8 % as operational dominance diminishes. However, these improvements assume Queensland achieves a substantial reduction from 0.79 to 0.26 kg CO₂eq/kWh by 2080. Currently, according to Australian Energy Statistics [102], electricity generation in New South Wales, Victoria, and Queensland relies heavily on fossil fuels, predominantly black coal and brown coal, highlighting the magnitude of required grid transition.

Moderate-emission grid cities show 1200–1500 kg CO₂eq/m² stratification. Melbourne (38.4 % reduction, 5993 to 3693 kg CO₂eq/m²), Sydney (35.4 % reduction, 5434 to 3512 kg CO₂eq/m²), and Perth (28.3 % reduction, 4451 to 3190 kg CO₂eq/m²) achieve substantial improvements, but face the dilemma where building performance depends critically on achieving projected emission factor reductions. These reduction benefits could disappear entirely if coal plant retirement

schedules are delayed or renewable deployment falls short of projections.

Low-emission grid convergence exposes optimization saturation limits. Hobart's minimal 7.1 % improvement despite further decarbonization reveals diminishing returns as electricity emissions approach zero. This raises critical questions about the cost-effectiveness of continuing building energy efficiency investments in increasingly decarbonized energy systems. Most concerning, Adelaide exhibits a 7.6 % performance degradation, with LCGHGE increasing from 2491.5 kg CO₂eq/m² in baseline conditions to 2680.6 kg CO₂eq/m² under the energy mix evolution scenario. This deterioration reflects fluctuating South Australian emission factors (Table S2), demonstrating significant uncertainty regarding future grid characteristics. The linear interpolation of emission factors between current and projected 2080 values ignores potential plateau effects and non-linear transition dynamics that could significantly modify LCGHGE. However, electricity system prediction extends beyond the scope of this research and represents an important avenue for future investigation. These findings underscore the inherent uncertainty in decarbonization pathways and energy mix transitions. While decarbonization appears promising, this variability highlights the risks of relying predominantly on external energy system transitions rather than demand-side measures.

The reliance on annual average emission factors represents a notable methodological simplification that warrants further examination. Dynamic grid emission factors, which capture hourly or sub-hourly fluctuations in electricity generation carbon intensity, have been shown to substantially alter building operational carbon assessments [43,81,82]. Amaripadath et al. [81] demonstrated that static emission factors can misestimate the operational carbon intensity of net zero energy buildings, with a 111 % underestimation in Paris and a 120 % overestimation in New York. Bertini et al. [43] similarly highlighted the need for dynamic modeling approaches when assessing whole-life carbon performance of building renovations, showing that using hourly emission factors can lead to a substantial increase in operational carbon emissions for ultra-low energy scenarios. In this study, the use of annual average emission factors likely underestimates the emission reduction potential of envelope designs that effectively manage peak loads, particularly in cooling-dominated tropical cities where peak cooling demand coincides with afternoon periods of higher grid carbon intensity. Conversely, in regions with high renewable penetration, midday solar generation may reduce grid carbon intensity precisely when cooling loads peak, potentially moderating the discrepancy between average and dynamic approaches. Incorporating hourly or seasonal dynamic grid emission factors, combined with projected future grid dispatch models, represents a critical avenue for advancing this framework and could reveal additional optimization opportunities through demand-side load shifting strategies that are invisible under average emission factor approaches.

Under baseline S1, operational emissions account for 74.2 % to 95.3 % of LCGHGE, creating optimization imperatives that prioritize thermal performance enhancement through increased material deployment. This operational dominance underscores the importance of implementing demand-side interventions, including high-efficiency HVAC systems and effective building retrofits, to address energy consumption at its source. Under S3, embodied carbon shares increase up to 27.8 % in Darwin. The observed proportional variations reflect the operational characteristics inherent to commercial office building typologies, where energy demands for equipment, heating, cooling and lighting typically dominate LCGHGE. For the analyzed 10-storey office building, embodied emissions remain substantially lower in magnitude compared to operational emissions. However, Robati et al. [70] reported a higher embodied proportion under grid decarbonization (58 %). This difference is attributable to several factors. Material quantities in this study are derived from a simplified BIM model excluding substructure, which likely underestimates absolute embodied emissions relative to as-built conditions. It is reported that LOD 200 models have been shown to yield GWP values 14.7 % lower than LOD 350/400 models [103], with

GWP increasing by 142 % when progressing from LOD 200 to LOD 350 [104], suggesting the absolute embodied values reported here are conservative. Differences in operational assumptions also contribute: Robati et al. [70] adopt lower internal heat gains (lighting: 3 W/m²; equipment: 5 W/m²) compared to this study (4.5 and 11 W/m², respectively) and a shorter study period (50 versus 60 years). Furthermore, more aggressive grid decarbonization is assumed in [70] (declining to approximately 0.2 kg CO₂eq/kWh by 2035), amplifying operational emission reductions beyond those projected in the present study. A cold climate study by Gaarder et al. [18] reported that when electricity emission factors exceed 25–30 g CO₂eq/kWh, operational GHG emissions dominate the LCGHGE in all case studies. High proportions of embodied impacts arise from highly efficient HVAC systems and energy self-sufficiency through local or onsite renewable energy generation [19,105], as well as cleaner energy mixes, making upfront carbon the dominant share of the total footprint. However, even when incorporating renewable systems in buildings, energy consumption should first be reduced to a reasonable range, as mismatches between energy consumption and renewable generation can be problematic [96], especially considering the intensified extreme events under climate change.

The findings of this study carry direct relevance for ongoing reforms in the Australian building sector. The NCC 2022 introduced strengthened energy efficiency requirements under Section J, raising the minimum thermal performance standards for commercial buildings. The results indicate that cities such as Townsville and Darwin, which exhibit the greatest sensitivity to both climate change and grid carbon intensity, warrant more stringent and climate-responsive envelope performance requirements than those prescribed under current national standards, which do not account for future dynamics. Furthermore, S3 results demonstrate that grid decarbonization delivers 28.32 %–38.37 % LCGHGE reductions in high-emission regions, benefits that are contingent upon achieving projected renewable energy transition targets under the Australian Energy Market Operator (AEMO) Integrated System Plan [106]. This interdependence between building performance and energy system evolution underscores the need for building policy to explicitly account for embodied carbon alongside operational performance under future conditions, consistent with the direction signaled by the Australian Government's Trajectory for Low Energy Buildings [107] and the NABERS embodied carbon rating framework [108]. The MTR analysis further suggests that carbon pricing mechanisms or embodied carbon disclosure requirements could serve as effective instruments for redirecting decarbonization investment toward cost-effective interventions in high-emission building stocks, rather than pursuing diminishing marginal gains in already high-performing buildings.

This analysis exposes critical assumptions about energy system transition pathways that may prove overly optimistic or simplistic. Predicting future energy mixes itself represents significant uncertainty arising from technological shifts, policy disruptions, and inconsistencies between climate pathways and energy system trajectories. Regarding technological shifts and policy disruptions, a Japanese study investigating future office building thermal performance illustrates similar uncertainties in energy system projections [109]. That study employed emission factors of 0.376 kg CO₂eq/kWh for 1990, with future factors rising to 0.551 and 0.689 kg CO₂eq/kWh, reflecting nuclear station accident impacts in Japan. The higher emissions scenario assumed 0 % nuclear power generation, while the lower estimate incorporated 20–22 % nuclear power capacity. Similarly, Robati et al. [70] considered different decarbonization strategies involving large-scale uptake of renewable energy, carbon capture technologies, and nuclear energy integration for energy mix in the Australian context, representing more radical transition pathways than those assumed in the present study. Regarding inconsistencies between climate and energy pathways, different climate models represent distinct decarbonization trajectories that could influence renewable energy production and the resulting emission factors. Su et al. [110] linked the Global Change Assessment Model to project electricity generation quantities under two levels of

policy implementation: a 450 and 550 ppm CO₂eq scenarios, incorporating different proportions of carbon capture and storage technologies and coal power. In a separate study [111], three scenarios of potential technological shift (energy saving, low carbon, and enhanced low carbon) were defined to predict the future energy mix in China. The first scenario assumed no significant technological breakthroughs in energy conservation or emission reduction, while the last assumed that key low-carbon technologies achieve greatly improved performance at low costs. In that study, 2010, 2020, 2035, and 2050 reference years were reported, with intermediate years calculated using linear interpolation, consistent with the approach adopted in the present study. While linear interpolation provides a tractable approximation, it necessarily masks potential non-linear transitions such as potential plateau effects, rapid technology deployment or policy reversals, especially over a 30-year interpolation span. Future research should explore scenario-based sensitivity analyses with alternative emission factor trajectories to better capture this uncertainty.

4.2.3. Dynamic climate and energy mix impacts on the embodied operational balance

Coupled climate-energy scenarios demonstrate substantial improvements over baseline conditions while revealing persistent climate burdens that compromise decarbonization benefits. S4 generally exhibits performance degradation relative to S3, with penalties ranging from 2.38 %–8.00 % for most cities (Fig. 7). Notable exceptions include Hobart (0.26 % improvement) and Canberra (0.17 % degradation) under SSP585.

Townsville exhibits full complexity: S4 achieves a 31.61 % reduction from baseline (8322 to 5692 kg CO₂eq/m²), substantially better than S1 but worse than S3's theoretical potential (5324 kg CO₂eq/m²) while showing 37.11 % improvement over S2's climate-degraded performance (9050 kg CO₂eq/m²). Grid decarbonization provides major benefits, but climate change imposes 6.89 % penalties (368 kg CO₂eq/m², S4 to S3). Brisbane demonstrates parallel patterns with a 31.52 % baseline improvement (6755 to 4626 kg CO₂eq/m²), with severe 7.93 % climate degradation under S2_SSP585 (7290 kg CO₂eq/m²). Darwin under

S4_SSP585 shows a 28.06 % baseline reduction (6618 to 4761 kg CO₂eq/m²), performing between S3 (4408 kg CO₂eq/m²) and S2 (7259 kg CO₂eq/m²).

Temperate zones reveal differential interaction patterns with reduced coupling effects. Melbourne S4 achieves a 36.71 % baseline improvement (5993 to 3793 kg CO₂eq/m²), closely approaching S3 performance (2.43–2.69 % degradation) while dramatically outperforming S2_SSP585 (6214 kg CO₂eq/m²). Sydney S4 SSP585 (Fig. 8) exhibits a 31.33 % baseline improvement, with 6.24 % climate-energy antagonism relative to S3 (3512 kg CO₂eq/m²). These 2.38 % to 6.24 % interaction effects suggest temperate climates experience weaker but non-negligible climate-energy coupling.

Importantly, Hobart and Canberra demonstrate near-convergence across S3&4, with Hobart showing minimal difference (1324 to 1321 kg CO₂eq/m²) and Canberra exhibiting similar variation (3582 to 3590 kg CO₂eq/m²), indicating negligible climate-energy interaction in heating-dominated systems. However, the apparent climate resilience in heating-dominated zones may prove illusory under more sophisticated climate impact modeling. The morphed weather files based on monthly climate averages may inadequately capture extreme events [37] and urban heat island effects [112], which significantly influence energy consumption and peak demand [12]. Furthermore, evidence suggests that people adapt to changing climate conditions, yet current modeling approaches may overlook occupants' adaptive capacity and behavioral responses that influence actual energy consumption patterns. Failure to account for dynamic occupant behavior and potential rebound effects could result in substantial over- or underestimation of actual energy consumption [93].

The performance divergence across scenarios exposes fundamental limitations in optimized design approaches that assume deterministic futures. Buildings optimized for S1 experience performance variations ranging from 10.90 % degradation in Adelaide (2491 to 2763 kg CO₂eq/m²) to 36.71 % improvement in Melbourne (5993 to 3793 kg CO₂eq/m²) under S4_SSP585. Designs optimized for S4 face up to 8.00 % thermal stress penalties under climate coupling relative to S3. This performance variability across plausible futures indicates that optimal design may be

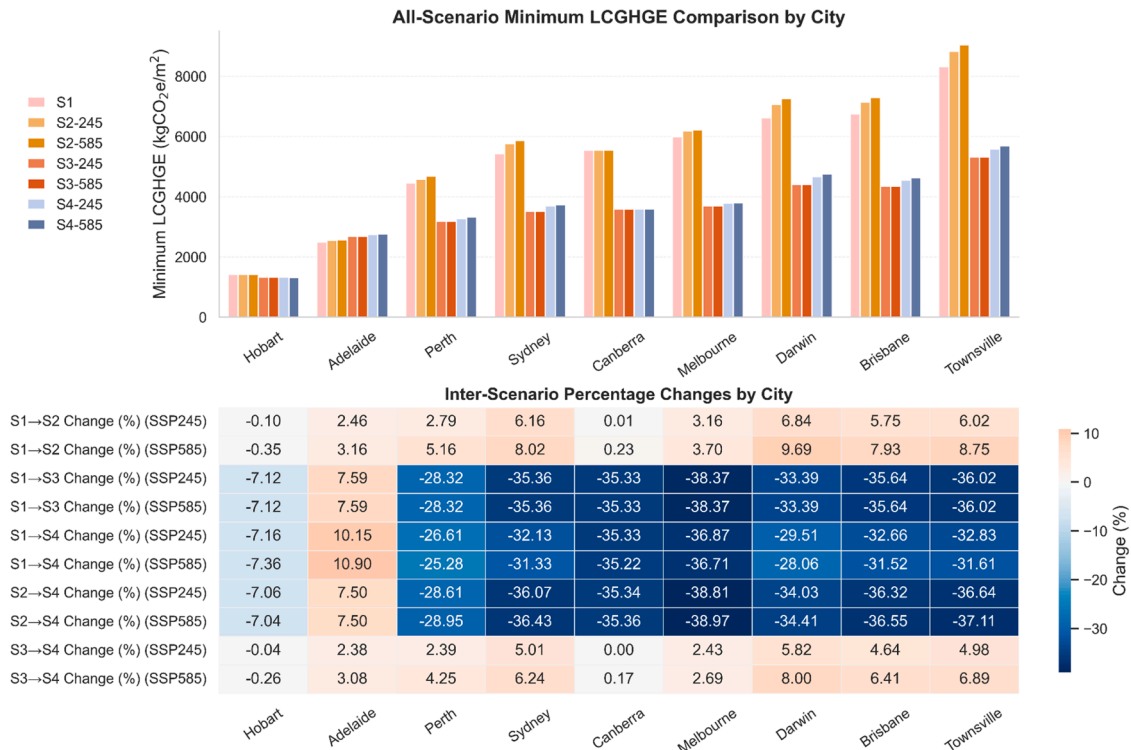


Fig. 7. All-scenario minimum LCGHGE comparison and inter-scenario percentage changes.

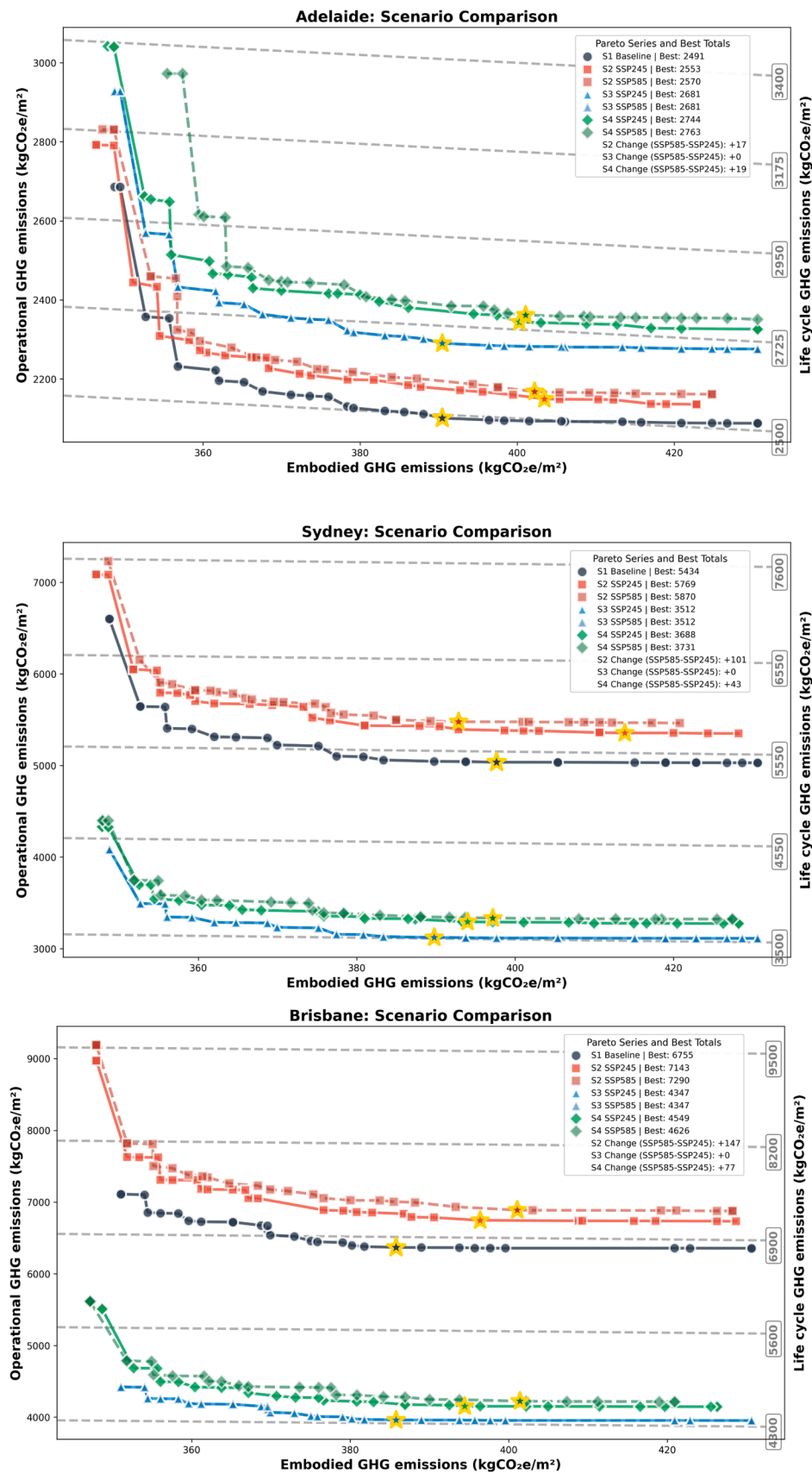


Fig. 8. Life cycle optimization comparison for Adelaide, Sydney and Brisbane.

fundamentally unachievable; pursuing performance maxima under any single scenario creates vulnerability to alternative trajectories.

Robust design strategies prioritizing consistent performance across scenarios necessarily sacrifice theoretical optima under certain conditions. Analysis of design parameter distributions across scenarios reveals different trade-offs: designs favoring lower embodied carbon (minimal insulation, higher WWR) perform well under S3 where operational penalties diminish, but experience severe degradation under S2 and S4 where climate impacts persist. Conversely, designs optimizing operational efficiency (increased insulation and optimized glazing) perform adequately under current and climate scenarios but face diminishing returns under grid decarbonization where embodied carbon becomes increasingly significant.

The cross-scenario performance matrices reveal significant design parameter sensitivities across different climate and energy system conditions. Optimal wall insulation thickness ranges demonstrate significant scenario-dependent variation: from 0.02 to 0.22 m (average 0.086 m) under baseline conditions, decreasing to 0.02 to 0.10 m (average 0.068 m) under S3 as embodied carbon penalties outweigh diminishing operational benefits. Conversely, these ranges expand to 0.02 to 0.40 m under S2_SSP585 (average 0.144 m) and 0.04 to 0.40 m (average 0.14 m) under S4_SSP585. Similarly, WWR values exhibit optimal performance at 0.20 across most locations. This is echoed in research by Gauch et al. [100], which reported that a lower WWR leads to lower thermal building energy for multi-storey buildings. The exception is Hobart, where the optimum range is 0.40 to 0.60 as glazing operational penalties diminish due to extremely low electricity emissions. However, designing buildings based solely on emission factors proves neither practical nor feasible in real-world applications. Building design practice must integrate multiple performance criteria, including economic considerations, daylighting requirements, occupant health and comfort factors, which may substantially influence optimal design parameters.

Robust design principles emerging from cross-scenario analysis include: (1) prioritizing thermal mass and adaptive envelope design strategies that provide climate resilience without excessive embodied penalties, as evidenced by consistent performance of moderate insulation levels across scenarios; (2) avoiding extreme design positions (minimal or maximal embodied/operational strategies) since optimal LCGHGE does not occur at these extremes, and these positions experience trade-offs between embodied and operational emissions; (3) integrating energy flexibility and demand response capabilities that leverage grid decarbonization while managing climate-driven peak loads, as operational emissions decline by up to 41.01 % relative to baseline under S3 (Melbourne).

5. Limitations and future research directions

This study establishes a methodological framework for evaluating embodied-operational trade-offs under dynamic climate and energy scenarios. Despite the efforts undertaken to ensure methodological rigor, this study has a number of limitations that should be clearly acknowledged and inform future research directions.

First, the envelope-focused parametric analysis constrains embodied carbon variation to 60–100 kg CO₂eq/m², while operational emissions vary by more than 2000 kg CO₂eq/m² (25 % to 40 % relative variation) in tropical climates. This 20:1 to 30:1 asymmetry in optimization impact suggests that envelope decisions primarily influence operational performance with marginal embodied consequences. However, grid decarbonization and renewable energy integration could invert this relationship. Structural material selections would significantly impact embodied GHG beyond the narrow range observed, such as the distinction between mass timber construction and conventional concrete and steel buildings. Future research should integrate structural material alternatives and construction method variations that could expand embodied carbon variability. Furthermore, the reference building was not calibrated against measured energy consumption data.

The use of standardized schedules and occupancy parameters based on NABERS assumptions ensures internal consistency but may not fully represent actual building operation, potentially affecting the absolute magnitude of operational emission estimates. Future studies employing measured data from comparable Australian office buildings would strengthen the empirical grounding of the framework. In addition, PV systems could substantially reduce operational energy consumption and associated emissions. When combined with BESS, on-site renewable generation can be complemented by strategic load shifting and demand response, further decoupling building electricity consumption from high carbon intensity grid periods. Future research should explore how such active carbon optimization strategies can be coupled with the LCA framework developed here, with explicit incorporation of the embodied impacts of these active system components.

Second, temporal characteristics of emissions present additional complexity not fully captured in this analysis. This study treats embodied and operational carbon as directly comparable within standard LCA frameworks, despite their different temporal profiles (immediate vs decades-long). Emissions occurring at different points in time can exert different climate forcing effects even when their cumulative totals are equivalent. Su et al. [111] employed time-dependent weighting factors to consider the different temporal global warming effects across scenarios. Beyond emission timing, predicting future energy system trajectories itself introduces substantial uncertainty. Technological shifts, policy disruptions, and inconsistencies between climate pathways and energy system evolution can produce divergent emission factor trajectories. Linear emission factor interpolation similarly masks potential non-linear transitions such as technology breakthroughs or policy reversals, particularly over extended 30-year interpolation spans. Therefore, future research should incorporate dynamic GWP weighting to differentiate near-term and long-term climate forcing, and adopt multiple energy system trajectory scenarios to avoid reliance on a single projected pathway.

Third, data resolution and modelling assumptions introduce further uncertainty. This study employs annual average electricity emission factors that do not capture variability in grid carbon intensity at hourly or sub-hourly resolution. Recent studies have demonstrated that such temporal granularity can substantially alter building carbon assessments [43,81,82], thus influencing the embodied-operational balance and representing a limitation that should be addressed in future studies. Furthermore, the time-dependent dynamics of biogenic carbon fluxes can significantly impact carbon footprint of mass timber construction [113–115]. For a 100-year analysis period, Dodoo et al. [114] revealed that static modeling of biogenic carbon yields 164 kg CO₂eq/m² for timber-frame buildings, whereas dynamic modeling with simplified biogenic carbon accounting produces 199 kg CO₂eq/m² (excluding biogenic carbon flows at year 100) and 91 kg CO₂eq/m² (including biogenic carbon flows at year 100). Andersen et al. [115] investigated the integration of forest management into building LCA and reported 6–81 % lower GHG emissions compared to the static method. Future research should address these uncertainties through hourly dynamic grid emission factors, dynamic biogenic carbon modeling, and sensitivity analyses examining the influence of reference study period, emission factor trajectories, and morphed weather data on trade-off outcomes. Given the prohibitive computational cost of NSGA-II optimization cycles, machine learning surrogate models may offer a viable pathway to accelerate such analyses.

Fourth, the carbon-focused optimization framework inadequately addresses multi-criteria decision-making requirements in real-world design practice. This raises concerns about overemphasizing carbon emissions reduction while neglecting broader social and material resource efficiency. For mass timber buildings, construction efficiency and productivity gains, which are not reflected in emission-focused indicators, remain unaccounted for in current assessment frameworks. Future frameworks should integrate economic indicators, thermal comfort assessment, and social equity analysis to support holistic

decision-making. Ultimately, advancing building sector decarbonization requires navigating coupled climate-energy-material systems at a broader urban scale.

6. Conclusion

This research introduces a multi-scenario LCA framework that integrates climate projections, energy system evolution, and multi-objective optimization to evaluate embodied-operational GHG trade-offs in building envelope design.

Through application of the framework to representative Australian office buildings, the study demonstrates profound spatial heterogeneity in building life cycle performance, with minimum LCGHGE varying six-fold across Australian climate zones, from 1426 kg CO₂eq/m² in cool temperate Hobart to 8322 kg CO₂eq/m² in tropical Townsville. This variation reveals that operational emissions dominate in conventional office buildings. Cities with high electricity emission factors exhibit substantial optimization potential through envelope enhancement, while low-emission regions demonstrate constrained performance variability and diminishing returns from thermal efficiency investments.

Climate change impacts degrade building performance across most Australian cities, with LCGHGE increases of 2.46 %–6.84 % under SSP245 and up to 9.69 % under SSP585. Tropical and subtropical regions experience the most pronounced performance penalties, with cooling-dominated climates facing operational energy increases that envelope optimization alone cannot adequately address. Conversely, heating-dominated regions like Hobart and Canberra demonstrate remarkable resilience, with minimal performance degradation even under severe climate scenarios.

Electricity emission factors remain one of the most critical factors in optimization. Specifically, S3 reveals substantial optimization potential, delivering 28.32 %–38.37 % LCGHGE reductions in high emission factor regions. However, these improvements are contingent upon achieving projected grid transition targets, exposing critical dependencies between building performance and external energy system evolution. High-emission, high-variability regions offer the greatest leverage for envelope-driven decarbonization and warrant concentrated regulatory attention. Low-emission grids approach optimization saturation limits, where continued efficiency investments yield progressively diminishing returns and may even prove counterproductive when embodied carbon considerations are incorporated.

This research challenges the prevailing paradigm of pursuing ever-lower operational energy consumption without considering embodied carbon implications and future energy system evolution. The optimization reveals that optimal solutions consistently occur at intermediate points along the embodied-operational trade-off, rather than at the extremes of either minimal embodied or minimal operational carbon. The MTR analysis quantifies efficiency thresholds beyond which additional embodied carbon investments fail to justify operational benefits. This framework provides practitioners with guidance for identifying optimization ranges where meaningful efficiency gains occur while avoiding counterproductive strategies that increase LCGHGE. Increasingly decarbonized electricity grids and expanded renewable energy integration would fundamentally invert traditional optimization hierarchies, potentially elevating historically minor factors (embodied) to dominance in the future. Such a transition underscores the growing importance of material efficiency strategies. The findings suggest that current building standards and policies would benefit from greater consideration of embodied impacts and dynamic factors, such as climate change and evolving energy mixes.

The framework's extensibility derives from three structural features that support future investigations. First, the four-scenario decomposition framework (static baseline, dynamic climate, dynamic grid and coupled conditions) can be applied to any building typology or scale without modification to its underlying logic. Second, the integration of open-access climate morphing tools, hybrid LCA methodology, and

parametric simulation enables straightforward substitution of input parameters, material datasets, and weather files to reflect different structural systems, building types or geographic contexts. Third, the MTR-based trade-off quantification approach is not specific to envelope optimization and can be extended to broader parameter sets, including structural material alternatives and active building systems.

The findings also provide practical implications for stakeholders. For building designers and architects, the MTR analysis identifies envelope optimization ranges where meaningful emissions reductions are achievable without incurring disproportionate embodied carbon penalties, providing actionable guidance for selecting insulation levels, glazing systems and WWR under both current and projected future conditions. For policymakers and regulators, the scenario decomposition demonstrates that building performance benchmarks based on static climate and grid conditions may not remain valid across a building's lifespan. Incorporating dynamic climate projections and evolving energy system emission factors into performance requirements would more accurately reflect LCGHGE outcomes. For building owners and developers, the results show that the relative value of envelope investment shifts substantially depending on future grid decarbonization trajectories. In high-emission-factor regions, envelope upgrades offer strong near-term returns; in grids undergoing rapid decarbonization, embodied carbon management warrants increasing attention as a complementary priority.

To sum up, this research establishes a methodological framework rather than prescribing universal optimal solutions, recognizing that embodied-operational trade-offs shift substantially across contexts. The four-scenario decomposition, separating static baseline, dynamic climate, dynamic grid, and coupled conditions, alongside the MTR-based trade-off quantification approach, provides a replicable structure that can be applied to other building typologies or geographic locations. The framework can also be extended to broader parameter sets including structural material alternatives and active or renewable building systems. By demonstrating that optimal LCGHGE is achieved at intermediate points along the embodied-operational trade-off frontier, this framework equips designers, building regulators, and policymakers with an evidence-based foundation for making climate-responsive, whole-life carbon decisions in an era of accelerating grid decarbonization and climate change.

CRedit authorship contribution statement

Zhuocheng Duan: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Hossein Omrany:** Writing – review & editing, Supervision. **Jian Zuo:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.buildenv.2026.114598](https://doi.org/10.1016/j.buildenv.2026.114598).

Data availability

No data was used for the research described in the article.

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